Retail Orders: Queries by Hand (Answers)

Fall 26

Calvin Williamson, Science and Math

Answers for the queries by hand handout. Each answer is followed by one SQL query that produces it. Other queries can give the same result.

The tables

customers

customer_id	first_name	last_name
1	John	Doe
2	Jane	Smith
3	Robert	Johnson
4	Sarah	Williams

products

product_id	name	price
1	Silk Blouse	89.99
2	Denim Jeans	79.99
3	Cashmere Sweater	149.99
4	Cotton T-Shirt	29.99
5	Wool Coat	199.99

orders

order_id	customer_id	product_id	order_date
1	1	3	2025-03-10
2	2	1	2025-03-12
3	3	4	2025-03-15
4	2	4	2025-03-18
5	1	2	2025-03-20
6	2	3	2025-03-25

1. Show the `first_name` and `last_name` of every customer who placed an order between 2025-03-15 and 2025-03-21. List each customer only once.

first_name	last_name
John	Doe
Robert	Johnson
Jane	Smith

```sql
SELECT DISTINCT c.first_name, c.last_name
FROM orders o
JOIN customers c ON c.customer_id = o.customer_id
WHERE o.order_date BETWEEN '2025-03-15' AND '2025-03-21';
```

2. For every order of a product that costs more than $80, show the customer's `first_name`, the product's `name`, and the `order_date`.

first_name	name	order_date
John	Cashmere Sweater	2025-03-10
Jane	Silk Blouse	2025-03-12
Jane	Cashmere Sweater	2025-03-25

```sql
SELECT c.first_name, p.name, o.order_date
FROM orders o
JOIN customers c ON c.customer_id = o.customer_id
JOIN products p  ON p.product_id  = o.product_id
WHERE p.price > 80;
```

3. For each product that was ordered at least once, show its `name` and `total_revenue`, its price times the number of times it was ordered, sorted from highest `total_revenue` to lowest.

name	total_revenue
Cashmere Sweater	299.98
Silk Blouse	89.99
Denim Jeans	79.99
Cotton T-Shirt	59.98

```sql
SELECT p.name, SUM(p.price) AS total_revenue
FROM orders o
JOIN products p ON p.product_id = o.product_id
GROUP BY p.name
ORDER BY total_revenue DESC;
```

4. Show the `first_name` and `num_orders` of every customer who placed more than one order.

first_name	num_orders
Jane	3
John	2

```sql
SELECT c.first_name, COUNT(*) AS num_orders
FROM orders o
JOIN customers c ON c.customer_id = o.customer_id
GROUP BY c.first_name
HAVING COUNT(*) > 1;
```

5. Show the `first_name` and `last_name` of every customer who has never ordered the Cotton T-Shirt.

first_name	last_name
John	Doe
Sarah	Williams

Sarah counts: she has no orders at all, so she never ordered the T-Shirt.

```sql
SELECT first_name, last_name
FROM customers
WHERE customer_id NOT IN (
  SELECT o.customer_id
  FROM orders o
  JOIN products p ON p.product_id = o.product_id
  WHERE p.name = 'Cotton T-Shirt'
);
```